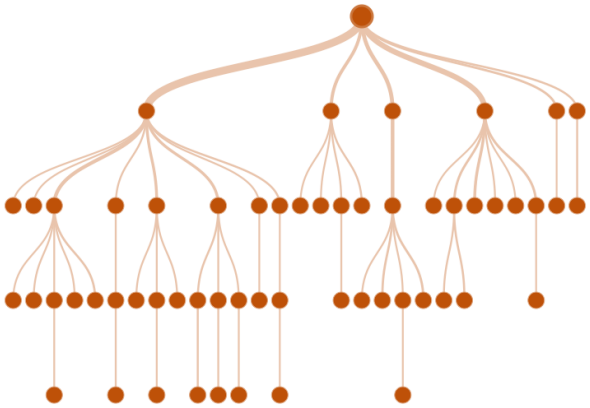
Creative Thinking for AI & IoT Innovations
Mid-semester in class test Question Bank
Submit PDF files only
Provide diagrams and pseudo code. Make necessary assumptions.

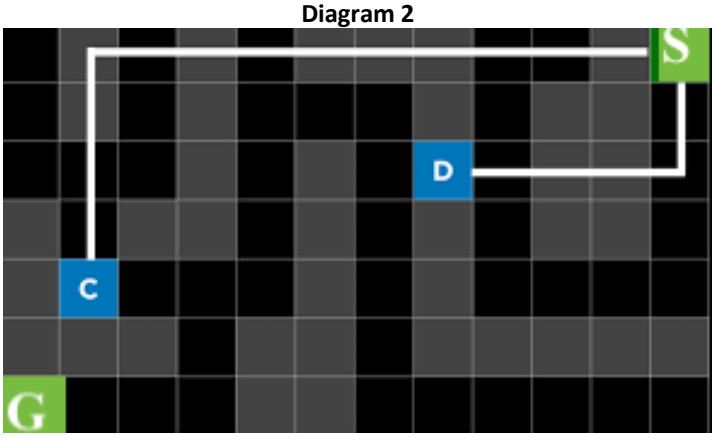
1. A person decides to use greedy best first search algorithm for website design. How it can be used in this case? What could be heuristic value in this case? Map node expansion and traversal to design changes. Discuss convergent and divergent approaches. Expand this approach to chat bot design with multiple trees. Can I represent every member of population as a decision tree (diagram 1)? How? Can heuristic function be dynamic? Write detailed step by step approach and pseudo code. Give one worked out example.

Diagram 1



2. For a product 'news reporting' you need to design goal-based agent with genetic algorithm as a part of agent function. S is a start state with original set of news. G is targeted news with reference impact expected on the readers. Every square in diagram 2 has fitness and it results from legal moves with genetic operators. Provide the detailed design of the agent. Take initial population (minimum four news), design 4 genetic operators, define fitness function. Give pseudo code and working algorithm for design with properly discussion about exit criteria. (Here no more than two black squares are allowed to be visited consecutively)

Diagram 2



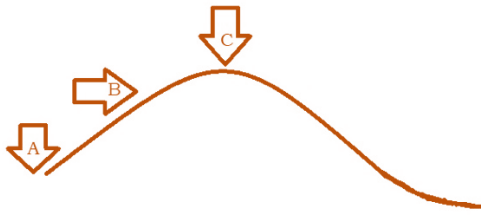
3. Airport authorities decide to use hill climbing algorithm for preparing and improving airport design and gate arrangements. The starting position of the design is defined based on blueprint of existing airport in another city. The goal state is decided based on voting by ten frequent travelers, average waiting time, optimal utilization of space and senior citizen friendly infrastructure. The legal actions for identifying neighbors are restricted to small change in architecture. Write pseudo code and explain working of the algorithm in the given scenario? What could be possible limitations in this case? What are local maxima in this case? (A starting state, B action selection, C goal state)

   Legal actions include:

   a. Adding an additional gate
   b. Increasing immigration counters by one
   c. Removing some unnecessary space
   d. Changing one shop position
   e. Reducing or increasing size of waiting area by certain units

4. What is common between best first search and hill climbing algorithm? How can I use best first search to expand the search space? Can it be possible to expand action space and search space simultaneously? Refer Diagram 3 – Can you control the convergence speed – can it be different at A, B and C – How? In Q3 define two additional operations to expand search space. What could be impact of it on the goal state?

Diagram 3



5. Discuss heuristic and non-heuristic search approaches. In product recommendation you want to use simulated annealing search algorithm to recommend products – how will you use it? Explain in detail with an example.

6. Design an intelligent agent to suggest creative modifications to the existing product. Think of using genetic algorithm and semi-supervised learning. What could be modifications possible to each of PEA and S to make an agent creative. Take a scenario of multi-agent systems where co-operative creativity is exhibited.

7. Is it possible to have dynamic heuristic values? Define heuristic value to search the best employee of the company and use hill climbing search to locate that employee. In this case how can you build a learning agent for this task where reinforcement machine learning and hill climbing work together. Make necessary assumptions.

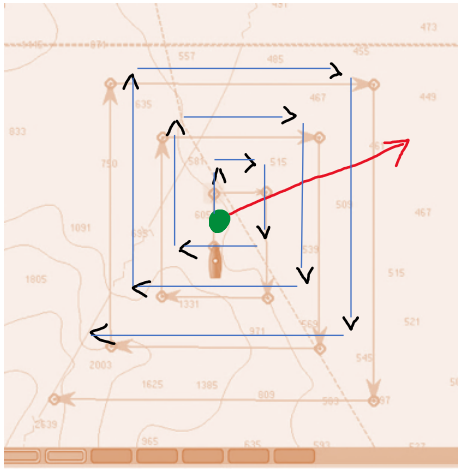
8. What is search space? Design a multi-agent system to combine two drawing patterns to draw new creative drawing by expanding search space or adding a new action. How can we combine two actions – explain. In this case both agents will introduce new combinations to create new states. What could be fitness function in this case?

9. Discuss different possibilities where psychological creativity is achieved through evolutionary algorithm. Can it be extended to historical creativity? What could be challenges in this case? Think of a few ideas to resolve these challenges.
   How can we expand conceptual space in this case?

10. Explain Intelligent agent and creative agent; intelligent search and creative search. In creative agent if you need to expand search space and action space how can you write a function for the same? Take an example of product design where you use utility based agent and reinforcement learning to expand action space. Explain different options.

11. Discuss connectionist and reductionist models. Think about use of Manhattan distance iterative deepening search (diagram 4) to expand the conceptual space. The next iteration expands search space. Think about using this approach to design features of the mobile phone. You may use certain paths for expansion as shown in figure below. Give pseudo code.

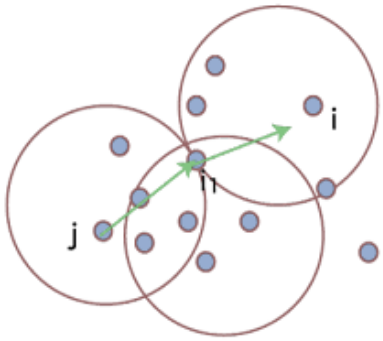
Diagram 4



12. In above example can we expand action space. In this case imagine that every action is combination of two basic actions. Define the basic actions in case of a particular game and expand the action space and use that algorithm to expand search space. The goal could be defined as a function of action and search space.

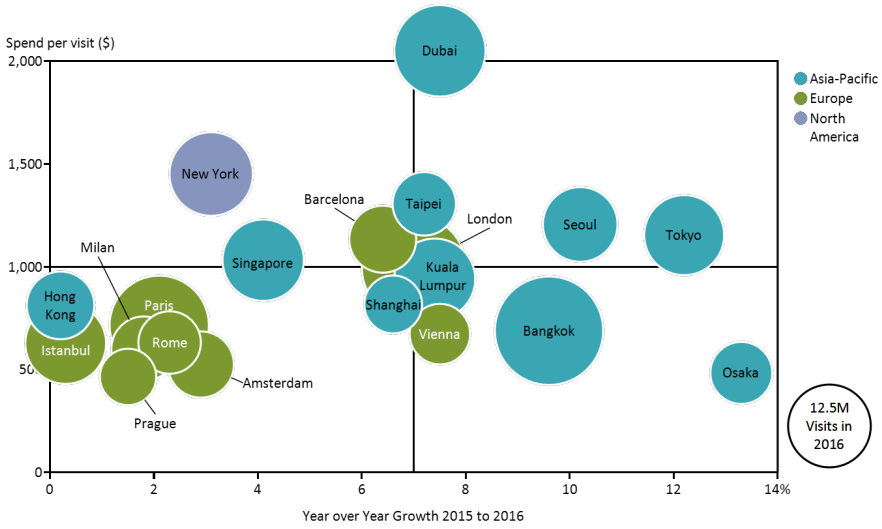
13. Model psychological creativity as a special function of historical creativity. In the beginning keep the constrained search space so the most of the actions could be creative. Slowly expand the search space and eliminate states – when no state is left in search space combine n top performing states and create new states. Continue the expansion till no new state created fall beyond the final expanded search space. Deliver final state set as the top ranked one. Refer diagram 5.

Diagram 5



14. Can we combine K-means clustering and KNN to develop semi-supervised learning. Can we use tourist destination clustering and organizing a tour from cultural perspective using this approach? How can we use combination of Euclidean distance and hamming distance in this case? How can you introduce systemic creativity in this case?
Refer diagram 6. Modify it to distribution based on cultural preferences.

Diagram 6



15. Model Product Recommendation using heuristic search algorithm.
    - Define heuristic function
    - Define Start State
    - Define Goal State
    - Define actions and neighboring states

Traverse search path for a specific example using hill climbing search (Assume 10 Product set to work out example). Discuss local maxima and global maxima in your case.

- How you can handle multiple goal scenario in this case?
- Provide a vector transition with correction using A* Algorithm
- Suggest interesting modification in hill climbing search algorithm that could lead to improved or interesting outcomes